Revenge of the Indifference Objection to Fine-Tuning

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Abstract

Indifference objections to the fine-tuning argument hold that the probability of fine-tuning on theism itself is very low or inscrutable, due to the vast range of alternatives God has for creating a life-permitting universe. We defend this response against a litany of recent objections. First, we develop a Bayesian model showing that one cannot relegate facts about alternative law structures into the background, since the very facts about those structures that make fine-tuning unlikely are themselves unlikely on theism in a way that cancels out any evidence from fine-tuning. Second, we point out that since God's options for producing a life-permitting universe always exceed the naturalistic possibilities, that we find ourselves in a universe with physical parameters counts as evidence *against* theism. Third, we show how our model addresses important defenses of the fine-tuning argument. For instance, we respond to the charge that certain objections to the fine-tuning argument 'prove too much', because it would follow from these objections that miracles such as the stars spelling out 'made by God' would be no evidence for theism. We show both that our model has unique resources to address this objection and further that this objection is independently mistaken.

Keywords Fine-tuning · Theism · Atheism · Bayesian Epistemology

1 Introduction

Bayesian fine-tuning arguments for theism proceed by arguing that the narrow range of life-permitting values in fundamental physics is more expected on theism than alternate hypotheses such as naturalism (Collins, 2009; Barnes, 2019). The key likelihood comparison is generally as follows: the probability of observing such a narrow life-permitting range given chance is astonishingly low, whereas on theism, it is substantially higher due to God's preference for a life-permitting universe.

According to indifference objections to fine-tuning arguments, fine-tuning is not evidence for theism because the probability that finely tuned constants will obtain on theism is itself very low or inscrutable. One way of developing this objection is by pointing out that God can achieve His desired outcome—conscious life—irrespective of how the parameters of physics are set. For instance, God could miraculously sustain conscious life by actively altering the matter and energy in the universe, use a different set of laws to govern the activity of conscious agents separately from the activity of all other matter, or create conscious beings in idealistic or supernatural

universes (Manson, 2018). If this is true and we have no reason to expect God to privilege physical life-permitting worlds over non-physical ones, then we lack any good reason to think that fine-tuning is more probable given theism than naturalism. Another indifference objection, from Sinhababu (2017), suggests that God could create morally valuable life by adjusting the psychophysical laws so that even primitive matter—such as electrons—could exhibit complex conscious states like love or empathy. Again, what seems to follow is that there is then no reason for God to favor our specific parameter values, since He could achieve His ends under different parameters by altering psychophysical laws. Finally, Weisberg (2010) argues that the fine-tuning data does not support a design hypothesis any more than life already did. Since God could have created 'lax' laws under which life arises across a wide range of constant values, then given life, it is no more likely by design than by non-design that we would end up in a world with fine-tuning.

Recently, indifference objections have come under attack. One reply holds that the very facts that make fine-tuning unlikely relative to alternatives—e.g., the psychophysical laws or law structures we observe—should be treated as background evidence when updating on the fine-tuning data, rendering them irrelevant to Bayesian fine-tuning arguments (Collins, 2009; Barnes, 2019; Donahue, 2024). A second reply presses a parity point: if theism is indifferent among life-permitting universes, then atheism should be equally indifferent, including among non-physical realms (Metcalf, 2022). A third objection, often aimed at fine-tuning critics more broadly, appeals to a companions-in-guilt argument: if the stars miraculously rearranged to spell out a verse from the Gospels, that would clearly confirm theism, and if indifference objections would also undercut arguments from such miracles, they must go wrong somewhere (Hawthorne & Isaacs, 2017; Barnes, 2019).

In what follows, we argue that these replies fail against the strongest forms of indifference objections. In §2 we develop a Bayesian model showing that placing our psychophysical laws into the background does not help the fine-tuning advocate: it merely relocates the problem, because those background facts are antecedently very unlikely on theism in a way that offsets the confirmation fine-tuning provides for theism. In §3 we argue the problem compounds once one considers God's non-physical options: the very fact that we inhabit a universe where conscious life is realized via physical parameters is itself more probable on naturalism than on theism. In §4 we answer objections to our model.

For the purposes of this piece, we take theism to be the claim that a morally perfect, omnipotent, and omniscient being—'God'—exists. We will also make three very generous assumptions for the fine-tuning proponent. First, if theism is true, it is epistemically certain that God will create life. Second, a restricted principle of indifference holds, on which equal-sized ranges of a natural variable are equiprobable. Finally, naturalism yields no informative constraints over the physical parameters relevant to fine-tuning. The fine-tuning proponent should be very happy with these assumptions, since the first, while likely false, strengthens their case significantly, and the second and third underpin Collins (2009)'s motivation for assigning a uniform prior over the values of the constants and initial conditions given naturalism.

2 Placing the Psychophysical Laws in the Background

The objection we will examine first targets Sinhababu (2017)'s argument that God could have instantiated conscious life under radically different psychophysical laws (henceforth 'PPLs'). According to the objection, this possibility is irrelevant to fine-tuning arguments, since our actual PPLs—those that attach consciousness only to complex organisms or functional states—can be held fixed in the background when judging the probability of fine-tuning. Once we conditionalize on our actual PPLs and update our priors for theism (*T*) and naturalism (*N*), we can proceed to run a Bayesian fine-tuning argument and find the following, where '*O*' is the observed fact that our universe falls within the life-permitting range of the constants and initial conditions, and "≫" denotes "is significantly greater than":

$$(1.) \quad \Pr(O \mid \text{PPL} \wedge T) \gg \Pr(O \mid \text{PPL} \wedge N)$$

Donahue (2024) proffers this response to Sinhababu's indifference objection, though this same point is relevant to most indifference objections: place a fact concerning our universe or its laws that narrows down the available life-permitting options for God in the background and evaluate the probability of fine-tuning given that fact. The objector can point out that this is arguably a basic component of diachronic updating: we consider each fact on its own, and after we have determined its probabilistic effect on naturalism and theism, we shift it to the background before considering subsequent facts.

In evaluating this move, a wrinkle should first be noted: If the background information in question includes only the actual laws of physics (minus the actual life-permitting values of the constants), then those laws tell us only what would happen in lieu of any supernatural causes. They say nothing about what might occur in possible universes with different values for the various constants given miraculous intervention by God. Thus, placing those laws in the background does not seem, by itself, to block indifference objections: for all the laws say, God could miraculously sustain conscious life regardless of how the constants turn out. At minimum, the background information would need to include the more modally robust claim that it is metaphysically possible only that a very narrow range of parameter values, relative to the total possible range, permit life. Given that God is omnipotent, it would be quite surprising if our physical laws made God unable to miraculously sustain agents in counterfactual conditions with alternate constants, and so it is not obvious that our background warrants this stronger claim that life is impossible given these alternatives on theism. Nevertheless, let us examine what would result from granting that the facts in our background do narrow down the life-permitting options available to God.

To understand the nature of our response, it will be useful to first explain and expand upon Weisberg (2012)'s objection to fine-tuning arguments. Weisberg points out that the laws governing the parameters for the constants and initial conditions are *stringent*, meaning that only a very narrow range of parameter values relative to the total possible range permits life. This stringency is precisely why our parameters are said to be fine-tuned and is what drives the purported evidential force of fine-tuning arguments. However, as Weisberg notes, God could have instantiated *lax* laws

instead—laws under which a much broader slice of parameter values relative to the total range could support life. Under lax laws, possible life-permitting universes are far more plentiful. An easy way to see this is visually, from Weisberg (2012):

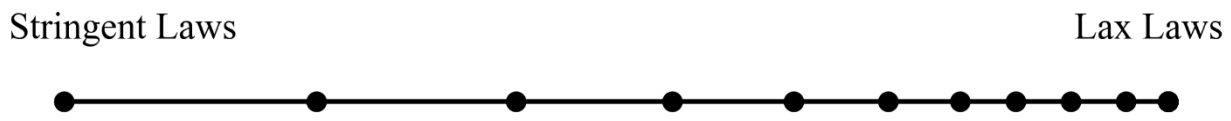


Figure 1. Distribution of lifeworlds across Law-Types.

We will call a *lifeworld* a possible universe with a law structure (sets of parameter values for the initial conditions and physical constants together with laws of nature) that supports conscious life. In Figure 1, the dots represent lifeworlds. More are clustered to the right, since lax laws allow for many more values of parameters that would permit life. If God will create life, He must choose one of these lifeworlds. However, we have no *antecedent* reason to think that God is more likely to choose one lifeworld over another, suggesting that each is equiprobable. To motivate this, we adopt the restricted principle of indifference from Collins (2009): *if we lack any reason to prefer one value of a natural variable p over another within a range R, we should assign equal epistemic probability to equal-sized ranges of p within R*. The purpose of the restriction to *natural* variables—those that occur in the simplest formulation of the relevant physics—is to avoid parameterization worries, such as Bertrand's paradox. To that end, we shall co-opt the use of 'natural variables' and revise it so it ranges over God's options in creating a universe. We will take a variable w, ranging over worlds God might create, to be 'natural' iff (i) w features in a natural description of God's act of creation and (ii) each admissible value of w uniquely picks out one possible world that satisfies God's stated aim.[1] Since God aims to create a lifeworld, the admissible values of w are precisely the lifeworlds. Let C be the epistemically illuminated parameter space of possible values for the constants and initial conditions, i.e., those values for which we can determine whether they are life-permitting.[2] Let R be the range of universes, each with a distinct law structure characterized by an ordered pair $\langle c,L \rangle$ where $c \in C$ and L is a law that fixes the width of the life-permitting region in C—lax laws corresponding to wider regions in C, and stringent laws corresponding to narrower ones. This suggests the following indifference assumption:

Divine-Indifference: Assign a uniform prior across all equal-sized sets of values for w within R. Treat each lifeworld within R that God might select as equally likely to obtain on theism.

[1] By 'natural description', we mean partitions that respect natural kinds, or otherwise do not involve any gerrymandered, grue-like predicates transformed to artificially privilege regions of the space of possibilities. While we are skeptical that there are natural representations in this sense, such a posit is required for the fine-tuning argument to avoid Bertrand paradoxes. We return to this issue in §4.2.

[2] Such a comparison range is precisely what Collins (2009) appeals to in his presentation of the fine-tuning argument to supply a non-arbitrary measure for the probability of fine-tuning on naturalism. A restriction to a bounded comparison range is needed to avoid measure-related problems (Colyvan et al., 2005).

On Divine-Indifference, God is *far* more likely to select a universe with lax laws, since such laws correspond to a much greater slice of lifeworlds. Conversely, on naturalism—and before conditioning on the existence of a life-permitting universe—we have, according to the fine-tuning advocate, no antecedent reason to favor lifeworlds over non-lifeworlds. As such, we also have no reason to expect lax laws over stringent ones. So, while stringent laws would be highly improbable on theism ex ante, they are not especially surprising on naturalism. This distinction is easily seen visually in Figure 1: if the points represent God's options, we are far more likely to end up in lax law space on theism, whereas on naturalism, neither lax nor stringent laws are more likely.

Let us now reconnect this to (1). Notice that an analogous move can be made in response to Weisberg's argument, letting S denote the stringency of our laws:

(1*.) $\Pr(O \mid T \wedge S) \gg \Pr(O \mid N \wedge S)$

However, (1*) alone does not license inferring:

(2.) $\Pr(T \mid O \wedge S) > \Pr(N \mid O \wedge S)$

If, in point of fact:

(3.) $\Pr(S \mid N) \gg \Pr(S \mid T)$

Placing our stringent laws in the background is therefore of no help here. That is, unless the truth of (1*) provides significantly stronger confirmation for theism *after* taking (4) into account. This is not the case, however. Given our assumptions, if all we know is that life exists, then the antecedent probability distribution over the set of lifeworlds with law structures varying in degrees of stringency or laxity would be equivalent conditional on theism or naturalism, giving us:

(4.) $\Pr(S \mid N \wedge O) = \Pr(S \mid T \wedge O)$

This holds because on theism, God creating life was already guaranteed, so if we learn only that life exists, that does not change our antecedent probability distribution over lifeworlds: Divine-Indifference would still apply. On naturalism, learning life exists only rules out non-lifeworlds, but it provides no reason for preferring one lifeworld over another. In both cases, we remain indifferent across the remaining options, giving us an identical, uniform distribution across lifeworlds. Although updating on the existence of a life-permitting universe initially confirms theism over naturalism, this shows that we do not get any further confirmation for theism upon learning that our law structure is stringent and thus highly unlikely to occur by chance. The overall lesson of this objection is summarized by David Manley:

> Even if our priors are such that the outcome supports theism when we *hold fixed the conditions that make the good outcome unlikely*, the outcome may not support theism when we do not hold those conditions fixed. When we ask whether theism *predicts those very conditions*, we may find that taking all our evidence into account does not support theism after all. (Manley, 2015, p. 6)

This point is also directly relevant to the objection that we ought to place our PPLs in the background when evaluating fine-tuning. For, what we know about PPLs is that they are incredibly *mind-unfriendly,* in the sense that only a tiny fraction of complex physical configurations can give rise to consciousness, and only a narrow range of parameter values in *C* can permit those configurations. In contrast, *mind-friendly* PPLs—such as those that Sinhababu (2017) proposes, on which rudimentary physical structures like electrons display complex, morally valuable mental states—impose far laxer constraints on *C*. For the same reasons previously adduced, there are far more lifeworlds under mind-friendly PPLs than under mind-unfriendly ones. To illustrate how this undercuts any attempt to shove these facts into the background, we provide a toy model:

Suppose we have a set of universes that each correspond to an element in our illuminated parameter space C, where each point in C fixes an equal-sized range of values for the fundamental constants. Let the cardinality of points in C be some finite cardinal 'x'. To simplify the calculation, suppose there are only two possible PPLs. One type of PPL is maximally mind-unfriendly (MU), so only one point in C permits life. The other is maximally mind-friendly (¬MU), so that every point in C is life-permitting. Each ordered pair of constant combinations and one of these psychophysical laws (e.g., ⟨c, MU⟩) maps one-to-one to a universe with that law structure. This gives us a cardinality of universes equal to the Cartesian product of C and the pair {MU, $\neg MF$} (2x). The relative evidence fine-tuning lends for theism is as follows, where 'O' again denotes a lifeworld existing:

$$\frac{\Pr(T \mid O \wedge MU)}{\Pr(N \mid O \wedge MU)} = \frac{\Pr(O \mid T, MU)}{\Pr(O \mid N, MU)} \times \frac{\Pr(MU \mid T)}{\Pr(MU \mid N)} \times \frac{\Pr(T)}{\Pr(N)}$$

Since there are x-many lifeworlds under mind-friendly laws and 1 lifeworld under mind-unfriendly laws, then given Divine-Indifference and that God will certainly choose a lifeworld:

$$(5.) \quad \Pr(MU \mid T) = \frac{1}{x+1}$$

On naturalism, if each universe is equally likely, whether it permits life or not, the parameter spaces for mind-friendly and mind-unfriendly universes both consist of sets of equal size, giving us:

$$(6.)\quad \Pr(MU \mid N) = \frac{1}{2}$$

Therefore, our Bayes factor (BF) based on MU is as follows:

$$(7.)\quad \frac{\Pr(MU \mid T)}{\Pr(MU \mid N)} = \frac{2}{x+1}$$

Recall we are also granting that $\Pr(O \mid T \wedge MU) = 1$ and $\Pr(O \mid N \wedge MU) = \frac{1}{x}$, which gives us a BF for the update on O of:

$$(8.)\quad \frac{\Pr(O \mid T \wedge MU)}{\Pr(O \mid N \wedge MU)} = x$$

With (7) and (8) in hand, and assuming neutral priors, we multiply $\frac{2}{x+1} \times x$ and observe that the posterior odds ratio for theism and naturalism, given both our universe falling in the life-permitting range and our laws being mind-unfriendly, is slightly less than 2:

$$(9.)\quad \frac{\Pr(T \mid O \wedge MU)}{\Pr(N \mid O \wedge MU)} = \frac{2x}{x+1}$$

Indeed, (9) is identical to the BF for Pr(O|T)/Pr(O|N), as can be computed via the Law of Total Probability:

$$\frac{\Pr(O \mid T)}{\Pr(O \mid N)}$$
$$= \frac{\Pr(O \mid T \wedge MU)\Pr(MU \mid T) + \Pr(O \mid T \wedge \neg MU)\Pr(\neg MU \mid T)}{\Pr(O \mid N \wedge MU)\Pr(MU \mid N) + \Pr(O \mid N \wedge \neg MU)\Pr(\neg MU \mid N)}$$
$$= \frac{1 \cdot \frac{1}{x+1} + 1 \cdot \frac{x}{x+1}}{\frac{1}{x} \cdot \frac{1}{2} + 1 \cdot \frac{1}{2}} = \frac{\frac{1+x}{x+1}}{\frac{1}{2x} + \frac{1}{2}} = \frac{\frac{1+x}{x+1}}{\frac{1+x}{2x}} = \frac{2x}{x+1}$$

Therefore, given generous assumptions, even with the massive BF in (8), the evidential impact provided by fine-tuning is roughly counterbalanced by (7). Of course, this model is based on idealized—and likely false—assumptions. For instance, the assumption that law structures are divided into just two 'maximally mind-friendly' and 'maximally mind-unfriendly' buckets is false.

Moreover, it may well be that the update on O purchases more than a BF of just under 2. Still, our conclusion does not depend on those simplifications. More generally, let $\alpha = Pr(MU|N)$, $\lambda = Pr(O|N \wedge MU)$, $\rho = Pr(O|N \wedge \neg MU)$, and $\beta = Pr(MU|T)$. In that case:[3]

$$(10.)\ BF(O \wedge MU) = \frac{\beta}{\alpha\lambda}, \text{ and } BF(O) = \frac{1}{\alpha\lambda+(1-\alpha)\rho}$$

Under Divine-Indifference, God has no antecedent preference among lifeworlds, so the probability that God actualized a MU lifeworld is simply equal to the proportion of lifeworlds that are mind-unfriendly. Letting $|LU|$ be the measure of MU lifeworlds and $|LF|$ be the measure of $\neg MU$ lifeworlds:

$$(11.)\quad \beta = \frac{|LU|}{|LU| + |LF|}$$

An uninformative prior over worlds conditional on naturalism will then assign each set of worlds a probability proportional to its measure. Hence, $\Pr(MU \wedge O \mid N) = \alpha\lambda \propto |LU|$ and $\Pr(\neg MU \wedge O \mid N) = (1-\alpha)\rho \propto |LF|$, which, with a common constant of proportionality, yields:

$$(12.)\quad \beta = \frac{\alpha\lambda}{\alpha\lambda + (1-\alpha)\rho}$$

Substituting (12.) into (10.) we have:

$$(12.)\ BF(O \wedge MU) = \frac{\frac{\alpha\lambda}{\alpha\lambda+(1-\alpha)\rho}}{\alpha\lambda} = \frac{1}{\alpha\lambda+(1-\alpha)\rho} = BF(O)$$

Put in plain language, the features that make O unlikely on naturalism, such as mind unfriendliness, are correspondingly antecedently unlikely on theism under Divine-Indifference. This is why shrinking the λ parameter, representing how unlikely fine-tuning is on N, cannot help the fine-tuning argument: β shrinks in exact proportion to it, and the improbability cancels out of the BF. Updating on $O \wedge MU$ is evidentially no better than updating on O alone.

3 Could the problem be worse?

Up until this point, the core of our argument has assumed that God selects among spaces of worlds with stringent or lax laws governing the physical constants, or mind-friendly or unfriendly laws governing psychophysical connections. However, there are far more options

[3] Assuming $Pr(O|T) = 1$—relaxing this only lowers the BF.

available to God that we have not discussed. In light of God's omnipotence, there is a vast ocean of possibilities within which God may create life without setting up parameters of physics, miraculously sustain minds regardless of the parameters, or select for law structures that give rise to conscious life with no corresponding physical substrate. As Manson (2018) points out, God might create a heaven where unembodied angels and spirits interact in harmony, create a universe where occasionalism or Berkeleyan idealism are true, or create a whole plethora of worlds that involve various supernatural interventions. On naturalism, however, no such non-physical possibilities exist. The set 'n' of lifeworlds possible on naturalism is a proper subset of the set 'm' of all possible lifeworlds. It is certain, on naturalism, that whatever the conditions are that allowed for the development of life, they had to be found in a naturalistic world picture. Given theism, however, God could have created any lifeworld within the total space of m, including non-naturalistic options. Moreover, there is no clear antecedent reason to expect God to prefer naturalistic lifeworlds over non-naturalistic ones. Formally, if D represents the datum that the parameters allowing for life fall within the range of n, then $\Pr(D \mid N \wedge O) = 1$, while $\Pr(D \mid T \wedge O) \approx \frac{n}{m}$, assuming a roughly uniform distribution across God's options.[4]

Thus, it is not just the stringency or mind-unfriendliness of the laws that drowns out the evidence fine-tuning provides for theism. The fact that these parameters are realizable in purely physical models constitutes further evidence for naturalism over theism. This suggests that if the bare fact that life exists counts as evidence for theism, even that support may be offset—or outweighed—by the further fact that such life depends on physical parameters and laws. Not implausibly, then, the total evidence relevant to fine-tuning, including that life exists at all—the fact that we have previously conceded favors theism—may be a wash, or even favor naturalism over theism.

A similar move could be made to the one we already examined: our background knowledge should include that our constants fall in the "epistemically illuminated" region of parameter space, in which contemporary physics allows for counterfactual modeling for changes to their values (Collins 2009; Barnes 2019). However, this maneuver fails for the same reason discussed in §2. The datum D concerning the physicality of our range of parameters is not evidentially neutral. Under N, D is guaranteed, whereas under T it is not. The lesson is that one cannot simply relegate D to our background knowledge and avoid our objections. After all, the total evidence is what *really* matters, and the epistemic probability of the constants hitting *our specific life-permitting values* within D under T depends on how likely D itself is conditional on T.

Metcalf (2022) objects that if, for all we know, God is indifferent between physical and non-physical realms, then the same should be true for whatever explains our world under

[4] One could object that God would not create morally impermissible worlds, so the denominator should be restricted to permissible lifeworlds. However, as we will argue later, this raises Pr(D|T∧O) only if the restriction to impermissible worlds disproportionately rules out more non-naturalistic options over naturalistic ones. Absent an argument for this claim, an appropriate indifference assumption should leave the ratio roughly unchanged. Furthermore, since naturalism entails D and theism does not, Pr(D|N∧O) = 1 > Pr(D|T∧O), and so D must disconfirm theism to some degree.

atheism—what he calls an "atheistic realm creator" (ARC). Since both God and an ARC select from the same space of possible worlds without regard for the realm type, the physicality of our world is no evidence against theism. Metcalf goes further, arguing that while any possible non-theistic world is compatible with an ARC, God is likely to have certain preferences and will not create worlds that conflict with them. Since physicality does not seem to conflict with God's preferences, our world being physical is in fact prima facie evidence *for* theism. We offer two replies.

First, the hypothesis we have assumed is not bare atheism but naturalism.[5] This is not an *ad hoc* move; not only is naturalism an independently motivated family of views, it is common practice to understand naturalism as the main competitor to theism for discussions on the fine-tuning argument. Therefore, the existence of possibilities for non-physical realms, given bare atheism, is irrelevant to our argument. Most accounts of naturalism will rule out such realms, as they often restrict reality to only fundamentally physical, non-mental entities. Metcalf objects that moving from the coarse-grained hypothesis of generic atheism to the more fine-grained hypothesis of naturalism sacrifices modesty, lowering its prior probability. That is true, but benign in this case, as theism itself is a highly fine-grained hypothesis when compared to bare atheism. The most generic competing hypotheses in this context are that the universe is designed or undesigned. However, neither hypothesis clearly yields more informative constraints for the kind of universe one should expect *a priori*. It is unsurprising that upon comparing theism—a specific design hypothesis positing an all-knowing, all-powerful, all-good being—to the most generic atheistic non-design hypothesis, theism wins on explanatory power. But this reasoning stacks the deck in favor of theism while ignoring the corresponding cost to *its* prior. A much fairer hypothesis test would pit theism against a more substantive atheistic hypothesis, such as naturalism. Absent an independent argument, we see no good reason to think theism is more modest or enjoys a higher prior than naturalism.

Second, while given theism, there are plausibly some worlds that are inconsistent with God's preferences; that fact does little for us if we don't know what those preferences are. Consider an analogy: you decide to eat at a fine-dining restaurant, "*Chef-O-Divine,"* where the chef selects a single entrée served to all guests each evening. Management assures diners of one thing: *"The chef never cooks anything that conflicts with his preferences."* You know some dishes are off the table—but unless you know the chef's tastes, you have no idea which dishes to rule out, nor any basis for expecting one dish over another. It would be intolerably arbitrary to assign greater probability mass to getting Filet Mignon over other dishes. Now suppose you might be at either Chef-O-Divine or another restaurant, Chef-O-Random, which randomly selects a dish from the same menu, but you do not know which restaurant you are in. Filet Mignon is not more expected at either Chef-O-Divine or Chef-O-Random, so receiving it provides no reason to increase your

[5] By naturalism, we broadly refer to the thesis that causal reality is coextensive with natural reality (Draper, 2017; Oppy, 2020). By natural, we refer to only those entities described by our best sciences, with the proviso that they are fundamentally non-mental. Whether naturalism is a substantive thesis to compare to theism depends on how these views are motivated, and we acknowledge this goes beyond our scope here.

credence in one restaurant over the other. This is true even though only Chef-O-Divine involves selection by preferences, because you are in the dark about the content of said preferences. One might argue that we do have some idea of what God's preferences are: He will actualize only those worlds it is morally permissible to create. Even so, this only shows that the probability that any universe God creates is morally permissible to create is 1. It does *not* follow that more non-physical lifeworlds in proportion to physical lifeworlds are excluded from the theistic outcome space relative to the atheistic outcome space. Thus, we still have no reason to believe that a uniform distribution over lifeworlds yields a higher likelihood of physical worlds existing under theism compared to atheism. To establish that physicality is evidence for theism, one must provide a principled measure under which physical worlds take up greater probability mass in the domain of God-compatible worlds than in the domain of atheism-compatible worlds. Until the required work is done to motivate such a measure, Metcalf's argument is deficient, even if it *were* to contact our objections.

4 Objections

4.1 Theistic Multiverse

A first objection charges Divine-Indifference with assuming that God must choose between individual lifeworlds when He might instead create multiple lifeworlds. Perhaps God will create a multiverse containing every possible lifeworld with its own laws, including stringent lifeworlds (Kraay, 2010). We make two points in reply. First, it is not the case that the existence of a multiverse undermines our calculations. It merely shifts the question from the probability that a lifeworld with stringent, mind-unfriendly laws obtains at all to the probability that we *observe* such a lifeworld. To accommodate this change, we can modify Divine-Indifference to the following:

Observation-Indifference: Treat each lifeworld within R as equally likely *to be observed* as any other given theism.

In the multiverse model under discussion, God brings every possible lifeworld into existence, and any of them can be observed. Thus, no observer-selection effect is at work. Further, as previously noted, universes governed by stringent laws like ours are extremely rare within the space of lifeworlds. Hence, given Observation-Indifference, the probability that we would find ourselves observing a universe with our laws remains astronomically low—with or without a multiverse. The presence of a multiverse no more affects that probability than knowing that there are many dice rolls affects the probability of observing a specific die roll "1" ten times in a row. The probability that we would *observe* a stringent universe like ours given multiverse theism, remains as follows, where |S| denotes the subset of lifeworlds with stringent laws and |LW| is the relevant domain of all lifeworlds:

$$\Pr(\text{ observe } S \mid T, \text{ multiverse }) = \frac{|S|}{|LW|}$$

Secondly, if the appeal to a multiverse could help, the move cuts both ways. Naturalists can invoke a multiverse to explain why we observe a fine-tuned universe. Indeed, multiverse models are among the most prevalent naturalistic responses to fine-tuning (Saad, 2024). If a sufficiently large multiverse can account for fine-tuning, then positing a designer plausibly becomes unnecessary.

4.2 Stringency Indifference

Another objection is suggested by White (2011)'s criticism of Weisberg's argument, refined by Dorst & Dorst (2022). White (2011) argues that one should not have greater confidence in design or non-design based on S (the stringency of the laws), because there is no reason to assume that a designer would favor stringent laws over lax laws. According to Dorst & Dorst, this suggests two potential indifference assumptions that we could use when modeling our expectations. One is Weisberg's life-indifference assumption, which is roughly equivalent to Divine-Indifference. Another, rival indifference assumption is uniform across levels of stringency, such as the following, letting a 'stringency class' be an equivalence class of lifeworlds defined by stringent or lax laws:

Stringency-Indifference: Assign a uniform prior across each level of stringency. Given theism, God selects within a range R* of stringency classes, each of which is equally likely.

The intuition behind this principle is this: Irrespective of how lax or stringent our laws are, God can intervene to set up the physical parameters and initial conditions to ensure that life comes about. God would therefore not care which stringency classes are used. Dorst & Dorst argue that we should have higher-order uncertainty about which indifference assumption is correct. Therefore, they propose that we take a weighted average of our credence across the defensible indifference measures, conditional on design & life and no-design & life. Under our assumptions, this will demonstrate that $\Pr(S \mid T \wedge O) \gg \Pr(S \mid N \wedge O)$. This is because Stringency-Indifference implies that while God is not unlikely to select stringent laws, they are incredibly unlikely given N∧O, since stringent laws entail that O is very unlikely to occur by chance. Conversely, Divine-Indifference simply implies $\Pr(S \mid T \wedge O) = \Pr(S \mid N \wedge O)$. Thus, if we average these two credence functions, we find that Stringency-Indifference contributes an overwhelmingly strong BF favoring theism when we condition on our total evidence. While averaging in Divine-Indifference does lower the total BF for theism, it does not come close to offsetting it. As a result, the total evidence still heavily favors theism. In the context of Sinhababu's indifference objection, the same point could be made: For, we could endorse a mind-friendliness indifference principle that is isomorphic with Stringency-Indifference, on which we assume that

God is indifferent across classes of mind-friendliness, as opposed to lifeworlds, or average our credence functions as Dorst & Dorst suggest. Either way, the total evidence will favor theism.

We begin by noting that this objection revives the specter of paradoxes of parameterization dependence, since Stringency-Indifference and Divine-Indifference are contradictory ways of parameterizing the very same outcome space. Recall that to avoid parameterization paradoxes, we endorsed a *restricted* version of the Principle of Indifference—one that permits only 'natural' parameterizations. If the fine-tuning proponent is to use a principle of indifference to motivate a uniform distribution over values of the constants, something like this restriction is needed. Without such a restriction, the principle collapses into incoherence, as the measure fails to remain invariant under nonlinear transformations. The stock illustration is from van Fraassen (1989). A cube factory produces cubes with side lengths between 0 and 2. What is the probability the next cube has a side length $s \leq 1$? If one assigns equal probability to the side-length cells $\{s \leq 1\}$ and $\{s > 1\}$, the answer is 1/2. However, the first cell corresponds to the face-area interval $A \in [0, 1]$. You might instead distribute evenly over face-area [0, 4], which divides into equal-width subintervals [0,1], [1,2], [2,3], [3,4]. This gives us a finer partition that yields 1/4 for $s \leq 1$. Finer still, one can reparametrize by volume $V = s^3 \in [0, 8]$, and get 1/8. However, Dorst & Dorst may suppose that we can circumvent this issue by taking a weighted average of our rival indifference assumptions. Indeed, because of how unlikely fine-tuning is, even if one assigns Stringency-Indifference an extremely small prior, one will receive a hefty update in favor of theism from the fine-tuning data. We now make several rejoinders to this argument.

One problem for this strategy is that if one can average over rival ways of partitioning the outcome space, such a move opens the door to infinitely many equally plausible credence functions to choose from. For example, one might assign a uniform prior over the possible masses of fundamental particles given theism, since, being omnipotent, God can always adjust other features to ensure life obtains regardless of the masses. Alternatively, consider an indifference measure over various types of fundamental properties that might be instantiated, or various possible distributions of such fundamental properties throughout spacetime, or perhaps across mathematical forms of the laws of physics, or classes of laws defined by algorithmic complexity (i.e., description length). In each of these cases, the rationale mirrors that of Stringency-Indifference: God can create life no matter which of these possibilities obtains, so He would be indifferent among these options. The problem compounds once we add that each of these parameterizations can be transformed in countless ways, by taking logarithms, inverses, squares, square roots, exponentials, or arctangent compressions of their various coordinates. Or we might consider larger sample spaces that include all of God's non-physical options, which themselves open a panoply of further possible parameterizations. Which of these partitions do we include and exclude? How do we weigh them? Even if one could offer principled answers to these questions and determine some well-defined credence measure, the prospects that this will vindicate the credences fine-tuning advocates hope for looks bleak.

A possible reply to this worry about averaging across credence functions is that while it is true that there are innumerable credence functions, some involve more intuitively natural

parameterizations than others.[6] Once we exclude parameterizations that are intolerably arbitrary,[7] Stringency-Indifference will be one of a limited set of major contenders. We offer four replies to this suggestion.

First, a stringency-class itself does not appear to be a natural property of universes, but simply one among an infinity of possible properties one might use to divide classes of lifeworlds. Natural properties are often justified by their explanatory power, and theoretical utility in supporting certain lawful generalities in our best scientific theories. 'Stringency-class' is not a theoretical term that appears in our best scientific theories. It does not seem that 'stringency-class' is the sort of property that could play the role of a reference magnet,[8] since it is a highly abstract, derivative classification artificially constructed for the purposes of dissecting fine-tuning arguments. Stringency classes appear to involve arbitrary disjunctions of universes, with no physically privileged commonality between them beyond the narrowness of their life-permitting band. Even if we relax our criterion for naturalness to countenance Stringency-Indifference, given that it is not a particularly natural property, it still seems as if there are innumerably many other parameterizations that are equally 'natural' candidates, like those mentioned previously.

Second, there seem to be equally natural measures that pull in the opposite direction: namely, that mind-unfriendly laws are even more unexpected on theism than what Divine-Indifference implies. Consider, for instance, Opportunity-Indifference: a credence function that applies indifference across possible opportunities for life to arise *within* universes. Since more mind-friendly laws permit even rudimentary physical structures to support life, such universes offer far more such opportunities—so Opportunity-Indifference assigns even greater probability density to mind-friendly universes than Divine-Indifference does.[9] Similar conclusions follow from other parameterizations: indifference over possible types of conscious life, degrees of fecundity in the production of life, or over all possible observers or civilizations. None of these are obviously grue-like, gerrymandered representations. In fact, given that Stringency-Indifference artificially bins together worlds, parameterizations over observers or replication events in which early life emerges are arguably more natural, as they pick out specific, concrete instances of natural phenomena. Hence, even if we restrict ourselves to a select few parameterizations, it is very unclear that averaging over them will yield the credences fine-tuning requires.

Third, insofar as one prefers more natural parameterizations, we believe Divine-Indifference is a better indifference assumption than Stringency-Indifference. One reason for this

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[6] We thank an anonymous reviewer for raising this point.

[7] For instance, representing a given law-structure in transformed coordinates by taking squares, logarithms, or similar functions.

[8] For Lewis, 'naturalness' is what makes certain properties better candidates for reference than others, as in his reply to Putnam's Model-Theoretic argument (Lewis, 1984). It also plays a broader role in his metaphysics in grounding intrinsic similarity, or Humean supervenience. Prima facie, stringency appears to satisfy none of these roles.

[9] Recall that we are considering Dorst & Dorst's reply in the context of Sinhababu's objection, so we ought to consider not only 'lax' laws but 'mind-friendly' laws of the kind described earlier. We point this out because it is easier to see how something like Opportunity-Indifference assigns greater probability density to universes with mind-friendly laws than lax laws.

is that Divine-Indifference appears to be less susceptible to disjunctive coarsenings, or refinements,[10] which are at the core of Bertrand-style paradoxes, as illustrated by the cube example. By laying out conditions (i) & (ii) for Divine-Indifference in §2, we offset potential concerns around such coarsenings and refinements, both by restricting ourselves to natural representations, and by stipulating that the admissible values of w must pick out individual possible worlds with life. Thus, for instance, any admissible transformations must *agree* on the property of uniquely picking out individual lifeworlds for credences to be distributed over. Any refinements that partition arbitrary segments or regions of lifeworlds, or coarsenings which parameterize over disjunctions of lifeworlds are inadmissible due to the stipulations in (i) and (ii). Together with the plausible suggestion that the identity conditions for worlds are not dependent on representation choices,[11] we go at least some ways towards offsetting parameterization worries.[12] By contrast, Stringency-Indifference parameterizes over classes of lifeworlds, and, as we argued, 'stringency' does not appear to be a natural property. Worse still, even if one has fixed levels of stringency as the parameter, there are many ways of defining stringency classes more coarsely or more finely that bin together more or fewer worlds—depending on the selected criteria for what counts as an equivalent 'level of stringency.'

Fourth and finally, even if there are a limited set of credence functions that pick out especially natural parameterizations, we are skeptical that we should restrict ourselves to these assumptions about the preference functions for God.[13] While grue-like parameterization choices look intuitively bizarre to us, it is doubtful that we are in a position to judge what would be a natural preference function from the perspective of a being so wildly different from us. Our judgments of naturalness usually depend on background knowledge implicit in theorizing about objects in the universe; there is little reason to expect them to apply to the psychology of a non-physical creator prior to creation itself. Furthermore, the rationale that Dorst & Dorst use for Stringency-Indifference can be applied here: no matter which decision procedure God uses to carve up the space of His options, life can be instantiated regardless. This holds whether the decision procedure he uses involves natural or unnatural representations.[14]

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[10] This terminology is borrowed from Norton (2008).

[11] If we understand a world as a maximal consistent set of propositions, e.g., any set S such that for any proposition P, either $P \in S$ or $\sim P \in S$, then the identity conditions of worlds should not depend on representation choices. Which propositions are elements of S are what fixes the identity of a given world. The same point applies if worlds are understood as the totality of spacetime and its contents, the identity conditions of a given world would instead be fixed by the total set of facts concerning the total distribution of properties across spacetime and any relations holding between them.

[12] Note that neither author *affirms* that these restrictions *in fact do* satisfyingly address Bertrand-style invariance arguments. However, to motivate assigning a vanishingly low prior to fine-tuning on naturalism something like this restriction is required, which is exactly what fine-tuning proponents have appealed to (Collins, 2009).

[13] We thank an anonymous reviewer for suggesting this argument.

[14] One could reply that we ourselves motivated Divine-Indifference via its purported naturalness. While this is true, it is the fine-tuning advocate that needs this restriction. If this objection is right and there is not *any* suitably natural parameterization, the fine-tuning argument cannot succeed. If there is a suitably

Putting aside our concern about averaging across indifference assumptions, there is another worry about Dorst & Dorst's proposal. There is a probabilistic tension[15] between the supposition that God will ensure that life obtains and the supposition that God is indifferent over degrees of stringency, and this casts doubt on Stringency-Indifference.[16] This is because the proportion of lifeworlds within law-space is inversely related to how stringent the laws are, meaning stringent laws correspond to far fewer possibilities for life. This is precisely what grounds the judgment that the stringency of our parameters is so surprising on naturalism *to begin with*, once we have life in our background. Whereas, if we were indifferent over stringency-classes, stringent parameters would not be particularly surprising. Prima facie, the same probabilistic tension appears to arise between the assumption that God will create a lifeworld, and His selecting a stringent universe, or being indifferent over stringency-classes. Put another way, if conditioning on life renders a stringency-indifferent credence function inadmissible given naturalism and life, some reason is owed for why that credence function remains admissible given theism — there must be a principled basis for the asymmetry in indifference assumptions. Absent independent grounds for thinking God selects among stringency classes, the very tension that powers the fine-tuning argument seems to equally tell against using Stringency-Indifference as one's rational credence function.

A final issue with Dorst & Dorst's strategy begins with the observation that, as they acknowledge, Stringency-Indifference is more supported on a broadly interventionist view of theism. This is because Stringency-Indifference implies that God is just as likely to select stringent laws as lax laws only because He will intervene to tweak the fundamental parameters of the universe to ensure life comes about. Yet, if that is the case, He should be equally willing to override life-hostile constant values themselves, e.g., by performing miracles that sustain life in an otherwise lethal cosmos, or by creating non-physical worlds that bypass physics altogether. This creates a thorny dilemma for the fine-tuning advocate. If we allow that God is an interventionist, we lose the motivation to think our specific constants are more probable on theism than on naturalism, *even given* stringent laws that narrow down God's *physical* options. That is because God will always intervene to secure conscious life regardless of the values of the constants; indeed, this is precisely what serves as the basis for indifference objections such as Manson (2018)'s. Conversely, if we adopt a non-interventionist model on which God avoids tinkering with chance-driven outcomes, this gives us a corresponding reason to prefer Divine-Indifference over

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natural one, we refer back to our previous arguments that Stringency-Indifference is less natural than Divine-Indifference.

[15] Probabilistic tension is a relation between propositions or events that is the converse of evidential support. If P is less probable given that Q, then given that ~Q, or given no information at all, P is in probabilistic tension with Q.

[16] A potential concern raised by an anonymous reviewer is that Stringency-Indifference, a normative principle about how to distribute credences, cannot be in tension with the descriptive fact that God is certain to create life. To clarify, this argument makes a bridging assumption: since the likelihoods conditional on theism are determined by our model of what God would prefer, we should adopt uniform credences over stringency classes only if our model of the preference function of God is indifferent over degrees of stringency, and there is tension between the descriptive propositions that God will certainly create life, and that God is indifferent over degrees of stringency.

Stringency-Indifference; God would be more likely to select law structures that bring life about without His intervention.

One might reply that God's intervention to set finely tuned parameters differs from an intervention to sustain conscious life. The former, one could argue, merely sets up the universe's initial conditions, whereas the latter involves disruptive miracles. This distinction falters once we consider the vast range of options available to God—many of which involve interventions on par with setting up the PPLs. For instance, God could intervene once to ensure that life operates under a different set of laws than those governing the physical universe. Such alternatives arguably even fit theism better antecedently. For instance, pleasure and pain could be tied directly to moral behavior rather than biological functions (Draper, 1989). Further, the stipulation that God would be just interventionist enough to fine-tune constants but not enough to bring about life in any other miraculous way looks *ad hoc* and tailored to retrofit theism to the data. Nevertheless, there is a kernel of insight here: if there is something uniquely valuable about our particular fine-tuned lifeworld, we may have reason to think God prefers it over alternatives. We now turn to that objection.

4.3 Fine-tuning Beyond Life

Yet another objection appeals to reasons beyond life that God might create our finely tuned universe. A popular, general strategy among fine-tuning advocates goes like this: While God has countless options for life-permitting universes, one can glimpse some reasons our world might be interesting to God—complex chemistry, spiral galaxies, etc. Even if each of these considerations carries little weight, so long as the probability that God would select our universe exceeds 1 in 10^{120}, the theistic hypothesis obtains a substantial probabilistic boost (Hawthorne & Isaacs, 2018). We believe this line of reasoning mistakes the mere act of incredulously pointing out the 1 in 10^{120} estimate of fine-tuning under naturalism—however vanishingly small that likelihood may be—for a good reason to think that the likelihood under theism must be greater. True enough, God might take interest in features of our universe—but equally, He might prefer features realizable only in universes radically unlike ours, such as worlds where inhabitants can rewrite causality, with Aristotelian cosmology, faster-than-light travel, dimensional travel, or time travel, and so on.[17] The possibilities are, quite literally, endless. Further, as we argued in §3, merely knowing that an agent has some preferences or other tells us nothing at all about which outcome they are more likely to choose, until we know what those preferences are or how they are weighted. The mere fact that God has some psychological tendencies and is reasons-responsive does not by itself substantially narrow His option space. Put differently, under the restricted principle of indifference, 1 in 10^{120} is not merely the prior assigned to our laws under naturalism; it is the

[17] Another way to see why this general point is insufficient is that it could have been levied in *any* lifeworld with a similar intuitive pull: 1 in 10^{120} is such a tiny number that it will always look intuitive that God has higher odds of preferring some feature that has obtained in said world than *that*. As we will argue, we believe this stems from a lack of imagination wherein one can easily point to interesting outcomes in our world, but one does not consider how the same outcomes (or alternate, more interesting outcomes) could obtain across a vast swath of God's alternatives.

default prior for *any* hypothesis until that hypothesis furnishes a reason to privilege the outcome we got. Worse yet, an omnipotent God has vastly more life-permitting options than are available on naturalism (see §2). Given a uniform prior over this vastly expanded menu, the probability that God creates a finely tuned physical universe like ours is plausibly *less* than 1 in 10^{120}—until we have an independent reason to think He prefers our lifeworld over incalculably many alternatives. In the absence of such a reason—or a principled measure that motivates the claimed likelihood ratio beyond brute intuition—this objection is unconvincing.

What if the theist can proffer some plausible reasons, apart from life-permitting conditions, that God is likely to select our fine-tuned universe? We do not object to this strategy in principle, but we are skeptical it can succeed for two reasons. First, there is a panoply of options God has for instantiating His desired outcome, or a similarly desirable outcome, sans fine-tuning.

For instance, suppose God desires scientific discovery—one suggestion for a reason God creates a fine-tuned universe (Collins, 2018). We deny that scientific discovery, or alternate modes of discovery of similar or greater value, are impossible in other worlds. Take worlds where God uses a separate set of laws to miraculously sustain life apart from physical matter. In these worlds, we can imagine all physical laws or processes behaving indistinguishably from the way they operate in our world. The same local regularities hold, experiments are repeatable, and nothing about scientific practice changes. The only difference is that there is a cosmological anomaly; the measured constants would prohibit life in the absence of miraculous sustainment by God. Alternatively, occasionalism might be true, so that nothing happens absent divine activity and even the laws themselves are just descriptions of the regular, uniform ways God chooses to act in the world. In this case, the whole world is supernaturally sustained, and yet science could proceed exactly as it does now by divine fiat. So long as these possibilities are not logically incoherent or *a priori* inconceivable, they prima facie clear the minimal bar of falling in the range of what an omnipotent being could actualize.[18] A further possibility is that God could grant persons in miraculous worlds the ability to both discover the laws of matter *and* the laws governing their consciousness and agency. Indeed, a universe with discoverable non-natural laws in addition to discoverable natural laws is plausibly *more* discoverable than ours, since it contains an additional law-governed domain open to investigation. Alternatively, in other miraculous worlds, God could allow persons to *simulate* universes and discover different laws across a vast variety of world conditions. Additionally, note that our own world appears to be substantially less discoverable than a host of interesting alternatives: for instance, in our world, history is only accessible in a very limited way, even through rigorous investigation. We are also fundamentally unable to understand or peer into the minds of others and fully see into their perspectives. Alternate worlds could therefore have allowed for many more interesting levels and kinds of discoverability, such as worlds with successful *supernatural* sciences including telepathy, psi, or clairvoyance, or non-scientific kinds of discovery such as those in which we have highly attuned, *a priori*, empathetic,

[18] One may raise doubts about the inference from conceivability to possibility because of modal skepticism (Van Inwagen 1998); however, such doubts plausibly undermine the kinds of modal claims needed to run the fine-tuning argument in the first place.

or moral-perceptual capacities that allow us to discover unique rational insights or relationship-centric goods. Importantly, God does not only have any one of these options, but a vast range of varieties for each of them, all of which potentially realize great goods that God may consider choice-worthy. Note also that so far, we have only considered immaterial or miraculous worlds. In non-miraculous material worlds with alternative psychophysical laws or laxer law structures, the degree of discoverability we have is not obviously compromised whatsoever. For this argument to succeed, what the fine-tuning proponent must show is that our specific outcome is *uniquely* and *optimally* discoverable within the expansive option set available to God—a demanding claim we doubt can succeed.

Similar points apply to the argument that God would desire embodied moral agents: Swinburne (2004) argues that embodiment is required for communication between persons, or for there to be clear consequences of one's actions. One problem with this proposal is that it evinces a lack of imagination. Consequences can take the shape of alternative laws applying only to conscious beings. Communication can happen by non-natural means, through direct telepathic connections between beings or some other supernatural route. More importantly, this only applies to a tiny slice of God's options for creating worlds without fine-tuning: in purely idealistic or spiritual worlds there may be no embodiment, but God could also sustain beings in physical worlds and simply use a different set of laws to govern their behavior than that of physical states in the universe, as we discuss in §3. And, of course, in worlds with lax or mind-friendly laws, there is no known metaphysical barrier that prevents God from creating embodied conscious agents. Once again, the fine-tuning proponent must show that there is something optimal and unique about our embodiment in this world, and we doubt this can be shown.

A second concern with this strategy is that the claim that the alternative goods supposedly enabled by fine-tuning are important to God is much less plausible than the original claim that God uses fine-tuning to support life. While we are skeptical of divine psychology generally, the argument that God would care about relationship-centric goods is one of the few goods that has been argued to be *a priori* derivable from God's attributes in-detail in the philosophy of religion (Schellenberg, 1993). However, when it comes to speculation on other kinds of goods enabled by fine-tuning, a healthy dose of skepticism is essential. For all we know, God might prefer opaque law structures and for the nature of the universe to be a mystery to us to cultivate a sense of humility and wonder. Alternatively, perhaps God would endow all creatures with innate knowledge that is conducive to their ideal flourishing without the need for discovery at all. Perhaps creatures being disembodied enables them to discover deeper truths about themselves or God, or to access a wealth of entirely different sorts of relationships with the world. Many such stories can be told from the comfort of an armchair, and when we confront the vast set of options God has, we doubt any specific story will be strongly grounded.

In sum, while there may be a fine-tuning argument for features other than life lurking in our future, we are skeptical of the tentative explanations given so far. What remains clear is that merely pointing out how improbable fine-tuning is *a priori* is insufficient: the defender of fine-tuning must do the hard work of ruling out God's many alternatives and arguing that the specific

goods associated with fine-tuning are especially likely relative to alternatives on theism. Evaluating every proposal that fits this case is beyond the scope of this piece, but we hope to have explained our general reasons for skepticism that such a strategy can succeed.

4.4 The Awesome Theistic Argument Test

Lastly, Hawthorne and Isaacs (2017) and Barnes (2019) propose the *Awesome Theistic Argument* (ATA) test for objections to fine-tuning. This test appeals to examples such as the following: Suppose tomorrow, stars in the sky rearrange themselves to spell out a verse from the Gospel of John. Intuitively, this arrangement would be powerful evidence for theism. Therefore, any objection to fine-tuning arguments that would equally undercut this argument cannot be right.

The ATA test might be levied against indifference objections as follows: God could have communicated the Gospel of John to us in innumerable ways. He may have written the Gospel of John in the genetic code of all living creatures, within every atom, or collectively beamed it into our heads, or used any number of unbounded means of communication within modal space. In light of indifference objections, we might infer that there is no good evidence for theism based on such miracles, since the probability that God would choose *any one* miracle in particular is astonishingly low. However, that is surely a mistake. We should accept that the ATA would be strong evidence for theism; therefore, indifference objections prove too much. This objection can be formulated as a companions-in-guilt argument:

(1.) If indifference objections defeat the fine-tuning argument, then they defeat the ATA.
(2.) It is not the case that indifference objections defeat the ATA.
(3.) Thus, it is not the case that indifference objections defeat the fine-tuning argument.

We will grant (2) for the sake of argument and instead dispute (1). Though no clear argument for (1) has been adduced by previous authors aside from the intuition of an analogy between these cases, we believe we can go beyond this lacuna and argue that there are symmetry breakers between the cases that show that (1) is unmotivated, if not false.

First, one can accept that the same reasoning present in indifference objections does apply to ATA cases, but in a way that still allows that the total evidence provided by ATA cases strongly confirms theism. Here's how: The total evidence provided by the ATA can be modeled as a two-step update. The first step involves updating on the disjunctive fact that some ATA-like miracle or other occurs, with each disjunct being an equally impressive miracle that would strongly support theism if actual. Examples may include the same message instead being encoded in our DNA, or the resurrection of all the dead bearing witness to God. The second step involves updating on the fact that, within the disjunction of ATA-like miracles, *this specific disjunct* occurred. An indifference objection to the ATA is successful *in the sense that* the second update on the specific disjunct does not contribute to the theistic case. It is indeed no more likely that God would choose *any specific* ATA-like miracle, such as the Gospel of John being spelled out in the stars, over any other. However, the ATA remains strong evidence for theism because the *general fact* that an

ATA-like miracle occurs is far more likely conditional on theism than on naturalism. Given theism, it is not terribly unlikely that God would use such a miracle to convey messages.[19] On naturalism, however, it is astronomically unlikely that any sort of ATA-like miracle will occur in general. Thus, updating on the disjunctive fact that an ATA-like miracle occurred knocks out a much greater chunk of naturalism's probability space. However, after renormalizing the probability space for theism and naturalism to versions of those hypotheses consistent with an ATA-like miracle, learning *which* miracle occurred is no further evidence for theism nor any corresponding blow to naturalism.

This reasoning parallels the indifference reasoning we defended: While the existence of life supports theism, *the specific way life arises*—via finely tuned parameters—adds no further confirmation for theism. Just as a fine-tuning skeptic is entitled to grant that the bare fact that there is life is evidence for theism, yet deny that evidence of fine-tuning strengthens that case, they are entitled to grant that an ATA-like miracle would be evidence for theism, even if its precise configuration does not strengthen this case. Therefore, if anything, the ATA demonstrates *why* indifference objections work, at least if one grants that life provides evidence for theism.

Second, there is a disanalogy between the ATA and the fine-tuning argument that casts doubt on the ATA test for objections proffered by fine-tuning skeptics more generally. Before discovering any message in the stars, we had ample background knowledge of how linguistic tokens are created. We know that intelligent agents use language and communicate specific messages, such that when they do, messages typically take the form of syntactically ordered symbols arranged in a specific rule-governed way that conveys semantic content. On the other hand, we have no background knowledge of such strings of linguistic symbols conveying intelligible semantic content arising from unguided physical processes or mere chance. Thus, we have overwhelming inductive support in our background knowledge ('kn') that such strings arise only through intentional action by agents with communicative goals. Indeed, given our relevant background knowledge, constitutive of what it is to treat something (e.g., a message in the stars) as a target for translation or interpretation, is, in effect, to regard it as a signal that features in communication among agents in our linguistic community.

Placing kn in the background allows us to treat scenarios such as those proposed in the ATA as *strong prima facie evidence* for agency. Plausibly, then, $\Pr(ATA \mid T \wedge kn) \gg \Pr(ATA \mid N \wedge kn)$. Were it the case that kn was not within our background knowledge, it is indeed doubtful that the ATA would supply significant evidence for theism. Suppose, for instance, we found ourselves in a world where our background knowledge, km, included frequently observed, specific messages that manifested because of well-understood stochastic physical events on cosmic scales. In such a world, we would develop an expectation to associate messages in the stars with completely unguided, chancy causes. We would, therefore, not be justified in accepting $\Pr(ATA \mid T \wedge km) \gg \Pr(ATA \mid N \wedge km)$.

A potential objection is that if we consistently observed messages with particular content, such as bible verses, woven into the fabric of the universe, this would intuitively be incredible

[19] Again, we assume this on behalf of the fine-tuning proponent.

evidence *for* theism![20] We must be cautious when considering this objection: finding *specific* messages, such as bible verses, may still be strong evidence for *particular* theisms only if we carry over our background knowledge that in *our* world, those verses are representative of certain religions. This is compatible with our argument: such specific messages are evidence for certain religions *only because* we have *separate* relevant background knowledge that undergirds that inference. However, if we had a plausible story for why syntactically ordered messages generally were the result of stochastic physical processes, we would no longer take the appearance of such messages generally as evidence for design, even if those messages being *weighted* toward specific religious claims may still be surprising. As an analogy, consider evolutionary theory: prior to having such a theory, our examples of complex structures emerging were the result of agency, so many people took complex organisms as evidence for design. The existence of a physical and unguided process that can produce such organisms undermines this inference, even if certain physical structures (e.g., watches) are still inferred to be the product of designers because we have *separate* information about their origins. Thus, this objection does not show that the evidential strength of the ATA is independent of our background knowledge, but instead that there are multiple potential sources of background knowledge that can support it. The overall lesson remains that for the ATA to buttress the fine-tuning argument, there must be analogous background knowledge in the fine-tuning case.

So far as we can tell, no remotely comparable inductive base exists for treating cosmological fine-tuning as the product of design. Nothing in our background knowledge gives us any evidence of agents creating universes, tweaking dimensionless constants, setting up early-universe boundary conditions, or dialing vacuum energies up and down. That is, unless we suppose the very point at issue: that a designer is responsible for such adjustments. Put another way: in the ATA cases, the inference to design is not based solely on the improbability of the configuration in question given naturalism, but on the fact that we have extensive precedent for intelligent agents producing exactly *this kind of* configuration. That precedent, plausibly, is what rescues the design inference from collapsing under the weight of the innumerable alternative possibilities we may distribute credence functions over. In the fine-tuning case, however, no such rescue is available.

Another possible objection to this argument is that we *do* have background knowledge pertaining to agents desiring to create life. Reproduction, after all, is an extremely common activity among humans and animals. We can thereby infer from this background knowledge that a designer is at least not terribly unlikely to create life. This response, however, is irrelevant to our argument, because the relevant data in our case is not that life exists, which we have already granted may be evidence for theism. The data that we are arguing is no further evidence for theism is *the stringency and fine-tuning of the constants that give rise to life*.[21] That we have a background expectation for agents to create life is therefore beside the point: what matters is whether we have good reason to

[20] We thank an anonymous reviewer for suggesting this objection.

[21] Recall that the point of many indifference objections is that life can be obtained without the need for fine-tuning.

think stringent, finely tuned parameters are more likely to be selected by agents than by naturalistic causes. For reasons already noted, there is no clear non-question-begging reason to believe our background evidence warrants this belief.

Finally, even in cases where the existence of a life-permitting universe is the relevant datum, this objection suffers from an additional difficulty. Every act of "life-creation" from agents that we have observed stems from biological reproduction, a contingent evolutionary adaptation that relies upon DNA, time, and certain environmental conditions. This replicatory machinery is the fundamental basis for the drive for reproduction in biological life. Consequently, we have no reason to expect such background information to apply to wholly non-physical, non-embodied, and non-biological life, which entirely lacks any such relevant machinery. By contrast, linguistic tokens and messages by their very nature plausibly carry an expectation of some communicative intent regardless of their medium—whether typewritten or inscribed in the stars. Thus, we have good reason to think that the cross-domain induction is justified in the latter case, but not the former.

5 Concluding Remarks

In this paper, we showed that indifference objections are more powerful and tricky than many proponents of fine-tuning claim. The same principles that saddle naturalism with an indifference assumption, and thereby motivate the fine-tuning argument to begin with, likewise saddle theism with an assumption that makes fine-tuning astronomically unlikely on theism. Attempting to avoid this conclusion by relegating facts about alternate law structures in the background is hopeless, because Bayesian analysis reveals that these facts are tied together and counteract one another. We also showed that our model is resistant to many defenses of the fine-tuning argument, in part because it captures the intuition that life itself may be evidence for theism, and that miracles may be evidence for theism, while allowing its adopters to remain fine-tuning skeptics. Thus, if the fine-tuning argument is to succeed, more work must be done to meet the challenge of indifference objections.

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